HORIZON EUROPE

Part B – Technical Description

AGRO-THRIVE Project Concept

AGRO-THRIVE

Call / topic	HORIZON-CL6-2026-01-CIRCBIO-09
Type of action	Research and Innovation Action (RIA)
Version	
Date	2026-07-30

Confidential

NOT SUBMISSION-READY - unresolved evidence blockers are listed in the evidence annex.

Part B — Technical Description

AGRO-THRIVE — Integrated governance and decision intelligence for balancing food security, bioeconomy, climate and biodiversity objectives

Call: HORIZON-CL6-2026-01-CIRCBIO-09 — Balancing food security, bioeconomy, climate and biodiversity objectives to unlock sustainable value chains · Horizon Europe · Destination: Circular economy and bioeconomy sectors · Call 01 – single stage (2026) · **Type of action:** Research and Innovation Action (RIA) · **Funding model:** Lump sum · **Duration:** 48 months · **Indicative EU contribution for the topic:** around EUR 6 million (WP 2026–27, Part 9, p. 185) · **Consortium:** **[INPUT NEEDED: consortium name]** · **Version:** **[INPUT NEEDED: proposal version]**

Page budget (working target 44 pages within the applicable limit; [INPUT NEEDED: confirmation of the applicable page limit and exclusions against the current official application template and the 2026 topic conditions]): §1 Excellence ≈ 17 pp · §2 Impact ≈ 10 pp · §3 Implementation ≈ 16 pp · figures and required tables included · 1 p rendering contingency. No annex is used; all required content is inside this document.

Evidence status. Requirements attributed to the call are cited as WP 2026–27, p. x (official Horizon Europe Work Programme 2026–2027, Part 9, pp. 185–187). Every statement requiring an external scientific, policy, market or statistical source is marked **[INPUT NEEDED: verified source with exact locator]** ; no external citation, comparative performance figure, maturity level, KPI value, budget or timeline is asserted without verified evidence or confirmed consortium input.

Table 1 — Participants (one legal entity per row; final numbering to follow Part A)

Participant no	Code	Legal name	Short name	Country	Organisation type	Role in AGRO-THRIVE
[INPUT NEEDED]	A	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	Coordinator; lead of WP1, WP2, WP10 (WP scopes to be confirmed, §3.1.3)
[INPUT NEEDED]	B	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	WP12 lead — cross-pilot synthesis, policy, replication
[INPUT NEEDED]	C	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: capability domain and task-level role]**
[INPUT NEEDED]	D	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: capability domain and task-level role]**
[INPUT NEEDED]	E	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: capability domain and task-level role]**
[INPUT NEEDED]	F	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: capability domain and task-level role]**
[INPUT NEEDED]	G	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	WP7, WP8 lead — regional multi-actor pilots

Participant no	Code	Legal name	Short name	Country	Organisation type	Role in AGRO-THRIVE
[INPUT NEEDED]	H	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: capability domain and task-level role]**
[INPUT NEEDED]	I	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: capability domain and task-level role]**
[INPUT NEEDED]	J	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: capability domain and task-level role]**
[INPUT NEEDED]	K	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	WP9 lead — business models, nature markets, value chains
[INPUT NEEDED]	L	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	WP3 lead — IARGF and stakeholder ecosystem
[INPUT NEEDED]	M	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	WP11 lead — dissemination, exploitation, communication, same-topic collaboration
[INPUT NEEDED]	N	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	WP5, WP6 lead — decision intelligence and actor-oriented services
[INPUT NEEDED]	O	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: capability domain and task-level role]**
[INPUT NEEDED]	P	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	WP4 lead — RBOS, data infrastructure, interoperability
[INPUT NEEDED]	**[INPUT NEEDED: the code "PI" used in earlier drafting is ambiguous]**	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	**[INPUT NEEDED: clarify whether this is a further beneficiary, an associated partner, an affiliated entity or a duplicate of code P, and state its role and participation status]**

Total number of beneficiaries, associated partners and affiliated entities: **[INPUT NEEDED]** (to be reconciled with Part A). The mandatory actor categories required by the topic — farmers, researchers, farm accountants, businesses (WP 2026–27, p. 187) — are mapped to confirmed participants in **Table 18 (§3.2)**. **[INPUT NEEDED: confirmation of the balanced multi-actor mix, with farmer, farm-accountancy and business beneficiaries or formally committed associated partners holding resourced tasks.]**

1 Excellence

1.1 Objectives and ambition

1.1.1 Challenge, opportunity and pertinence to the topic

European agricultural systems must simultaneously deliver food and feed security, supply biomass to a growing circular bioeconomy, contribute to climate mitigation and adaptation, and protect and restore biodiversity. These objectives compete for the same finite land, biomass, water and farm labour, yet the decisions that allocate these resources — taken by farmers, businesses, regional authorities and policymakers — are typically taken within sectoral mandates and without systematic treatment of trade-offs, leakage effects or cumulative impacts **[INPUT NEEDED: verified sources with exact locators characterising fragmentation of agricultural resource governance and uncoordinated trade-off management]** . The topic addresses precisely this imbalance, requiring projects to reconcile market incentives for bioeconomy and nature markets with the safeguarding of food supply and the environment, while enabling farmers to diversify income (WP 2026–27, pp. 185–187).

Farmers are at the centre of this tension. They are expected to supply food, feed and biomass, deliver environmental stewardship and adopt new business models, yet their bargaining position, access to decision-grade information and remuneration for environmental services remain weakly supported by current instruments **[INPUT NEEDED: verified sources on farmer bargaining power, fair-value distribution and remuneration for ecosystem services]** . The topic therefore demands more than analysis: tools that support farm income, farm management and natural-capital management; business models and value chains delivering fair value; and governance approaches co-created with farmers, researchers, farm accountants and businesses.

AGRO-THRIVE responds with an integrated, multi-actor governance and decision-support ecosystem: an **Integrated Agricultural Resource Governance Framework (IARGF)**, a **Regional Bioeconomy Operating System (RBOS)**, an **Advanced Decision Intelligence Engine** with actor-oriented services, and **farmer-centred business models and value-chain configurations**, co-created, demonstrated and validated in regional multi-actor pilot hubs. The design covers the topic's full scope: balancing competing objectives; applying natural-capital accounting to real decisions; using residues, secondary streams and underutilised biomass and land; assessing all four food-security dimensions including leakage and water conflicts; and delivering evidence-based policy and business recommendations (WP 2026–27, pp. 186–187).

1.1.2 General objective

> **General objective.** AGRO-THRIVE will enable coordinated, adaptive and evidence-based governance of agricultural resources by delivering and validating an integrated, multi-actor governance and decision-support ecosystem that reconciles food security, bioeconomy development, climate action and biodiversity protection in real regional settings.

Project success is defined as end-of-project changes attributable to project outputs, not as delivery of documents: (i) regional authorities and actor groups in the pilot territories apply IARGF rules to at least one real contested resource decision each; (ii) farmers, advisers and farm accountants use the services on real holdings and accept them against pre-agreed criteria; (iii) at least one co-designed business-model configuration per pilot is validated by the actors who would operate it; (iv) policy recommendations are traceable to identified evidence and delivered into named decision processes. **Baselines, target values and acceptance criteria for each: [INPUT NEEDED: measurable project-level success criteria with baselines, targets and due months, to be fixed once the consortium, pilots and resources are confirmed.]**

1.1.3 Specific objectives

SO1 — Develop and validate the IARGF. The governance backbone: principles, system boundaries, actor roles, safeguards, thresholds, evidence requirements and decision rules for allocating agricultural resources among competing food, bioeconomy, climate and biodiversity uses. Pertinence: the topic requires governance approaches that balance market incentives while safeguarding food supply and the environment, and that anticipate and manage trade-offs (CR1, CR2).

SO2 — Architect and deploy the RBOS. An interoperable evidence infrastructure connecting relevant data, models, tools and analytical workflows, with explicit reuse of existing European and regional assets to avoid duplication. Pertinence: capitalisation on existing research and tools, complementarity with related EU initiatives and accounting frameworks (CR11, CR12).

SO3 — Develop and verify the Advanced Decision Intelligence Engine. Analytical capability to evaluate alternative resource pathways from farm to macro level across economic, social, biodiversity and climate dimensions, with binding admissibility gates for food security, ecological integrity, food/feed safety, quality and regulatory suitability applied before any optimisation. Pertinence: interdisciplinary farm-to-macro research, supply–demand and price-threshold analysis, land-use conflict assessment across all four food-security dimensions (CR6, CR8, CR9).

SO4 — Co-develop innovative business models and sustainable value-chain configurations. Analysis of existing models and validated farmer-centred new models, including cascading and non-extractive configurations for residues, secondary streams and underutilised biomass and land, tested for profitability, accessibility, fair-value distribution and consumer acceptance. Pertinence: new business models delivering fair value and income to farmers without compromising food supply (CR4, CR5).

SO5 — Co-create, demonstrate, validate and transfer the ecosystem. Pilot evidence, territorially adapted action plans, governance roadmaps, policy recommendations and a replication package, produced with farmers, researchers, farm accountants and businesses throughout. Pertinence: mandatory multi-actor approach, effective SSH contributions, policy recommendations enabling replication across diverse European regions (CR10, CR13, CR14).

Table 2 — Objectives, verifiable results, KPIs and verification

Obj.	Verifiable result	Key performance indicators (baseline → target)	Measurement & verification	Owner*
SO1	**R1** Validated IARGF (principles, boundaries, roles, safeguards, thresholds, decision rules), released under version control	Framework applied to real contested decisions; acceptance by actor category — baseline [INPUT NEEDED]; target **[INPUT NEEDED: number of documented applications, acceptance thresholds per actor group, instruments]**	Independent methodological review (§1.2.11); documented application in pilot decisions; actor-signed validation statements	WP3 lead L
SO2	**R2** Interoperable RBOS integrating data, models, tools and workflows, with provenance and reproducibility	Connected assets; reuse-versus-build ratio; reproducibility and service performance — **[INPUT NEEDED: interoperability standards, connected-asset count, performance criteria, service levels]**	Interface and interoperability test records; reproducibility tests; provenance logs; asset-reuse register	WP4 lead P

Obj.	Verifiable result	Key performance indicators (baseline → target)	Measurement & verification	Owner*
SO3	**R3** Verified Decision Intelligence Engine, admissibility gates and actor-oriented services	Scenario, gate, validation and usability coverage — **[INPUT NEEDED: scenarios assessed across all four food-security dimensions, leakage and water; model-validation, uncertainty, explainability and usability targets]**	Benchmark, sensitivity and regression tests; structural or empirical validation records; task-based user validation	WP5/ WP6 lead N
SO4	**R4** Portfolio of analysed existing and validated new farmer-centred business models and value-chain configurations, each with a transfer-condition statement	Models analysed / co-designed / validated; fair-value and accessibility criteria met — **[INPUT NEEDED: counts, criteria, thresholds]**	Farm-accountancy and market assessment; actor validation records; price-threshold analysis	WP9 lead K
SO5	**R5** Pilot evidence base, regional action plans, governance roadmaps, policy recommendations, replication package	Pilot completion against acceptance criteria; recommendations traced to evidence; committed replication users — **[INPUT NEEDED: final pilot count, acceptance criteria, adopter and transferability targets]**	Cross-pilot validation report; evidence-traceability audit (gate G8); stakeholder endorsement	WP7/ WP8 lead G; WP12 lead B

*Partner codes pending legal confirmation of the consortium (Table 1).

Figure 1 — AGRO-THRIVE concept and objective–result architecture (placeholder, max 0.5 page): single integrated diagram showing IARGF, RBOS, Decision Intelligence Engine, actor services, business models and pilot hubs as one ecosystem, with arrows mapped to SO1–SO5, R1–R5 and the four expected outcomes. **[INPUT NEEDED: figure production.]**

1.1.4 State of the art and advances beyond the state of the art

AGRO-THRIVE's positioning is comparative by design: each of six innovations addresses a specific limitation of current governance frameworks, digital infrastructures, decision-support tools, safeguard practice, natural-capital accounting and business models. The limitations stated below are the project's working premises, to be tested in WP3–WP5; the named comparators, maturity baselines and measurable advances required to substantiate a beyond-state-of-the-art claim are marked as inputs and will be inserted with exact locators before submission. No comparative performance claim is asserted beyond what verified sources support.

Table 3 — State of the art, innovation advance and maturity

Innovation	State of the art (named comparators)	Limitation addressed (working premise)	AGRO-THRIVE advance	Measurable advantage	Maturity: baseline artefact → target functionality/ environment	Validation environment	Owner*	Key risk
I1 IARGF	**[INPUT NEEDED: named verified comparator frameworks]**	Fragmented sectoral governance; sustainability boundaries rarely operationalised into decision rules	Operational rules linking decision authority, evidence requirements, safeguards, trade-off procedure, accountability and territorial adaptation	**[INPUT NEEDED: metric and target]**	**[INPUT NEEDED: documented baseline artefact and current validation]** → **[INPUT NEEDED: target functionality and demonstration environment]**	Regional multi-actor pilot hubs	WP3 lead L	R1, R9
I2 RBOS	**[INPUT NEEDED: named verified platforms/ data spaces]**	Fragmented datasets, models and domain tools; limited technical, semantic and organisational interoperability	Orchestration of data, models, services and governance workflows across three interoperability layers, reusing existing assets	[INPUT NEEDED]	[INPUT NEEDED] → [INPUT NEEDED]	Integration testbed, then pilot hubs	WP4 lead P	R2, R4
I3 Decision Intelligence Engine	**[INPUT NEEDED: named verified decision-support/ modelling tools]**	Environmental, economic, social, behavioural and governance dimensions rarely integrated in one farm-to-macro chain	Coupled multi-scale scenario evaluation with thresholds, uncertainty, synergies, trade-offs and ranked, explained options	[INPUT NEEDED]	[INPUT NEEDED] → [INPUT NEEDED]	Technical verification, then pilot demonstration	WP5 lead N	R3, R4

Innovation	State of the art (named comparators)	Limitation addressed (working premise)	AGRO-THRIVE advance	Measurable advantage	Maturity: baseline artefact → target functionality/environment	Validation environment	Owner*	Key risk
I4 Safeguard-constrained decision logic	**[INPUT NEEDED: named verified safeguard/assessment approaches]**	Food security, ecology, food/feed safety, quality and regulatory suitability assessed separately, if at all	Binding admissibility gates classifying pathways by food security, ecological integrity, leakage, safety, traceability and compliance before optimisation, with logged rejections	[INPUT NEEDED]	[INPUT NEEDED] → [INPUT NEEDED]	Pilot hubs with qualified safety and regulatory experts	WP3 lead L; WP5 lead N; **[INPUT NEEDED: food/feed safety and regulatory owner]**	R1, R5
I5 Natural-capital and ecosystem-service accounting for decisions	**[INPUT NEEDED: named verified accounting/valuation frameworks]**	Accounts and valuations weakly connected to farm, business, governance and policy decisions	Accounts translated into gate thresholds, pathway comparisons, revenue/cost lines and governance choices, articulated with the Economic Accounts for Agriculture and farm accountancy	[INPUT NEEDED]	[INPUT NEEDED] → [INPUT NEEDED]	Pilot farms, accounts and regional decisions	**[INPUT NEEDED: natural-capital scientific lead and responsible partner]**	R3
I6 Farmer-centred business models and regional value chains	**[INPUT NEEDED: named verified comparator models/cases]**	Weak farmer bargaining power, unclear fair-value distribution, limited stewardship remuneration	Cascading and non-extractive configurations combining biomass use, ecosystem-service remuneration, conditional nature-market mechanisms, aggregation models and fair-value analysis	[INPUT NEEDED]	[INPUT NEEDED] → [INPUT NEEDED]	Pilot value chains with market operators	WP9 lead K	R6, R8

*Partner codes pending legal confirmation. TRL values are deliberately not stated; maturity is expressed as baseline artefact → target functionality and demonstration environment, and will be completed only against documented baselines (§1.1.5).

1.1.5 Ambition and R&I maturity

The ambition lies in **integration with enforceable safeguards**, not in any single component. Where current practice addresses food security, climate, biodiversity and bioeconomy objectives through separate instruments, AGRO-THRIVE couples: (i) governance rules that make trade-offs explicit and accountable; (ii) an interoperable evidence infrastructure that reuses rather than rebuilds; (iii) decision intelligence that compares only pathways which have passed binding food-security, ecological, safety and legal gates; (iv) natural-capital accounting that reaches actual farm, business and policy decisions; (v) business models tested for fair value under realistic market and transaction-cost conditions. Each advance is individually verifiable; their combination in one co-created ecosystem, demonstrated in real regional multi-actor settings, is the claim beyond the state of the art.

Scope is deliberately sized for a **Research and Innovation Action** of 48 months at the topic's indicative EU contribution of around EUR 6 million (WP 2026–27, p. 185): components progress from documented baseline artefacts and prototype-level elements to an integrated ecosystem demonstrated in relevant real-world multi-actor environments. Two mechanisms protect feasibility: (a) a **reuse-first rule** — no dataset, model or tool is built where a fit-for-purpose asset can be reused, adapted or interfaced (§1.2.4); (b) a **minimum coherent model chain** — only model components required by a confirmed pilot decision are developed or coupled, with all other components excluded from scope (§1.2.5, gate G1). AGRO-THRIVE does not claim market transformation within the project lifetime; it commits to validated frameworks, verified tools, pilot evidence, transferable roadmaps and evidence-based recommendations. **[INPUT NEEDED: maturity definitions and documented evidence for every proposed maturity progression in Table 3.]**

1.1.6 Alignment with the topic: expected outcomes and scope coverage

Table 4 — Contribution of objectives to the topic's expected outcomes (WP 2026–27, p. 186)

Expected outcome (topic)	Contributing objectives → results	Principal mechanism
EO-1 Decision-makers have improved understanding of bioeconomy and nature-market impacts on agricultural sustainability, food security, climate, biodiversity and land-use conflicts	SO1, SO2, SO3, SO5 → R1, R2, R3, R5	Farm-to-macro scenario evidence with leakage, cumulative-effect and water-conflict analysis, delivered through authority-facing services and regional action plans
EO-2 Society benefits from economic activities aligning bioeconomy, climate and biodiversity objectives while safeguarding food security	SO1, SO3, SO4, SO5 → R1, R3, R4, R5	Safeguard-constrained admissible pathways; validated value chains; safe-circularity screening of residues and secondary streams
EO-3 Farmers gain opportunities to diversify production and income and improve environmental performance without compromising food supply	SO3, SO4, SO5 → R3, R4, R5	Farmer and adviser services; farmer-centred business models with fair-value assessment; natural-capital remuneration pathways with verification-cost transparency
EO-4 Policymakers are better equipped to develop more effective, evidence-based agricultural and environmental policies	SO1, SO5 → R1, R5	Policy-to-requirement traceability matrix; logged pathway rejections as policy evidence; governance roadmaps and replication package

Table 5 — Call scope and requirement register (CR) — one atomic requirement per row; the register is the project's compliance backbone and is re-verified at gate G8. **[INPUT NEEDED: exact atomic quotation and precise locator for each row, extracted from the final official topic text.]**

ID	Requirement (topic)	Locator	Proposal response	WP / task	Result
CR1	Balance market incentives for bioeconomy and nature markets while safeguarding food supply and the environment	WP 2026–27, pp. 185–186	IARGF decision rules + binding admissibility gates before optimisation (§1.2.3, §1.2.6)	WP3, WP5	R1, R3
CR2	Anticipate and manage trade-offs, synergies and conflicts of use	pp. 185–186	Non-dominated option sets, trade-off governance procedure, rejection logs (§1.2.5, §1.2.6)	WP3, WP5	R1, R3
CR3	Apply natural-capital and ecosystem-service accounting to real decisions	p. 186	Six-step accounting chain feeding gates, comparisons, business models, monitoring (§1.2.7)	WP4, WP5, WP9	R2, R3, R4
CR4	Analyse existing and develop new business models delivering fair value and income to farmers	pp. 186–187	Baseline analysis, co-design, assessment, validation with transfer conditions (§1.2.8)	WP9	R4
CR5	Valorise residues, secondary streams and low-value or underutilised biomass and land	p. 186	Safe-circularity gate, cascading and non-extractive configurations, sustainable-availability baselines (§1.2.6, §1.2.8)	WP5, WP9	R3, R4
CR6	Interdisciplinary farm-to-macro research on economic, social, biodiversity and climate impacts	p. 186	Coupled model chain with coupling contracts and behavioural layer (§1.2.5)	WP5	R3
CR7	Provide tools supporting farm income, farm management and natural-capital management	pp. 186–187	Actor-oriented services with explainability, uncertainty and accessibility rules (§1.2.9)	WP6	R3
CR8	Assess biomass supply, demand and market-price threshold effects	p. 186	Scenario architecture separating drivers from choices; price-threshold switching analysis (§1.2.5, §1.2.8)	WP5, WP9	R3, R4
CR9	Assess land-use conflicts and all four food-security dimensions, including leakage and water	p. 186	Gates 1–3 with dimension-specific tests, leakage boundaries, water-conflict assessment (§1.2.6)	WP5	R3
CR10	Deliver evidence-based policy and business recommendations enabling replication	pp. 186–187	Policy traceability matrix, governance roadmaps, replication package (§2.1, §3.1)	WP12	R5
CR11	Capitalise on existing research and tools; avoid duplication	p. 187	Asset-reuse register with recorded reuse decision and non-duplication statement (§1.2.4)	WP4	R2

ID	Requirement (topic)	Locator	Proposal response	WP / task	Result
CR12	Create complementarities with related EU initiatives and accounting frameworks (incl. EAA)	p. 187	Named-initiative complementarity map; EAA articulation in the accounting boundary (§1.2.7, §2.2.3)	WP4, WP9, WP11	R2, R4
CR13	Apply the multi-actor approach with a balanced mix of farmers, researchers, farm accountants and businesses	p. 187	Four-level multi-actor design with governance rights and influence records (§1.2.12, Table 18)	all WPs	R1–R5
CR14	Ensure effective contributions from social sciences and humanities	p. 187	Four embedded SSH strands with defined integration points and review rights (§1.2.13, Table 10)	WP3, WP5, WP7–WP9, WP12	R1, R3, R4
CR15	Undertake collaboration with projects funded under the same topic	p. 187	Dedicated resourced task T11.4 with joint workshops and common communication and dissemination activities (§2.2.3)	WP11	R5
CR16	**[INPUT NEEDED: any further atomic scope, eligibility or destination-level condition identified in the final official topic text and general an - nexes]**	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]	[INPUT NEEDED]

Policy alignment is treated as a traceability obligation, not a claim: each relevant provision of the EU Vision for Agriculture and Food, the CAP, the EU bioeconomy strategy, the Carbon Removals and Carbon Farming (CRCF) Regulation and European Green Deal climate and biodiversity objectives is linked in T3.1 to a concrete framework rule, indicator or decision test **[INPUT NEEDED: exact legal and policy titles, versions and locators to be traced; number of provisions targeted]** .

1.2 Methodology

Method names are design commitments, not claims of demonstrated superiority. WP and task identifiers indicate where a method is executed; the work plan itself is in §3.1 and is not repeated here. Concepts are defined once and cross-referenced.

1.2.1 Methodological rationale and intervention logic

The topic is defined by simultaneity: the same hectare, biomass flow, water body and farm labour hour are claimed at once by food and feed production, bio-based value chains, climate mitigation and carbon-removal schemes, biodiversity restoration and emerging nature-market instruments (WP 2026–27, pp. 185–186). The call asks not for a better single-purpose tool but for the capacity to **balance** objectives from holding to macro level (pp. 186–187). If competing claims are analysed sequentially by separate disciplines and tools, the trade-off is resolved implicitly. AGRO-THRIVE therefore treats the balancing act itself as the research object and organises all methods around one explicit, auditable **decision problem**, instantiated per region and use case:

> Given a defined territory, resource base, actor constellation and policy environment: which allocations of land, biomass, labour and capital are **admissible** under food-security, ecological, safety and legal constraints; among the admissible allocations, which are **preferable** under economic, social, climate and biodiversity criteria; how **robust** is that ranking to uncertainty; and under which institutional, market and behavioural conditions can the preferred allocations actually be **implemented** by the actors who must implement them?

Four design premises structure §1.2. **P1 Admissibility precedes optimisation** — food security, ecological integrity, food/feed safety and legality are binding gates, not objectives tradable against income (§1.2.6; CR1, CR9). **P2 Trade-offs are shown, not aggregated away** — the default output is a set of non-dominated options with consequences, distributional effects and uncertainties visible; weights, where used, are elicited, versioned and varied (§1.2.5). **P3 Evidence must be actionable by named actors** — analysis is complete only when it reaches a decision taken by a farmer, accountant, cooperative, business or authority (§1.2.9; CR7). **P4 Co-production is a method** — actors define the problem, set requirements, co-design, host pilots, validate and co-author recommendations (§1.2.12; CR13).

Intervention logic — nine movements, two cycles. (1) Co-define regional decisions, actors, boundaries and safeguards; (2) assess and reuse existing data, models, tools and policy knowledge; (3) build the IARGF and the RBOS foundation; (4) develop and verify the Engine; (5) convert analytical capability into actor services; (6) co-design business models and value chains; (7) demonstrate and validate in regional multi-actor pilot hubs; (8) refine and **retest** changed components; (9) synthesise cross-pilot evidence into action plans, policy recommendations and replication roadmaps. Movements 3–8 are traversed **twice**, in two formally gated cycles (Cycle 1: gates G1–G4; Cycle 2: gates G5–G8, §1.2.11 and §3.1.2), so pilot evidence changes the framework and models rather than merely illustrating them. **[INPUT NEEDED: cycle start and end months, release plan and freeze dates, to be fixed in the Gantt (Figure 5) once resources are confirmed.]**

Figure 2 — Methodological workflow, information flows and feedback cycles (placeholder, max 0.75 page; single integrated figure replacing all narrative flow diagrams). Five layers with explicit flows: **L1 Co-definition (WP3, WP7)** — actor/institutional mapping → regional decision problems → boundaries, safeguards, indicators, decision rules → requirement set to all downstream layers. **L2 Evidence foundation (WP4)** — asset-reuse register → data architecture and governance → technical/semantic/organisational interoperability → traceable, versioned workflows. **L3 Analysis (WP5)** — scenario design → coupled farm-to-macro chain → admissibility gates → multi-objective and multi-criteria comparison → uncertainty and robustness. **L4 Services and configurations (WP6, WP9)** — actor services; business models, financing and value-chain configurations. **L5 Pilots and synthesis (WP7, WP8, WP12)** — baseline → deployment → 13-dimension validation → refinement → cross-pilot synthesis → policy recommendations and replication package. **Loop A** (within pilot cycle): validation findings → component refinement → retest. **Loop B** (once per cycle): cross-pilot learning → revision of IARGF rules, safeguards and thresholds → model re-verification. **Cross-cutting bands:** multi-actor governance; SSH; gender and inclusion; open science and FAIR management; quality gates and research integrity.

1.2.2 Concepts, system boundaries and assumptions

Operational definitions first. A version-controlled, published project glossary and indicator dictionary (T3.1) fixes at minimum: agricultural resource system; decision space versus context space; admissibility; safeguard versus criterion; natural capital, ecosystem-service flow, condition; residue, secondary stream, low-value and underutilised biomass and land; cascading use; non-extractive configuration; fair value; leakage; and the four food-security dimensions (availability, accessibility, utilisation, stability). Definitions are adopted from authoritative policy, legal and statistical frameworks wherever one exists, rather than invented **[INPUT NEEDED: authoritative definitional sources with exact locators, including the frameworks adopted for food-security dimensions, natural-capital accounting and biomass/residue classification]**.

Boundaries are declared per analysis, not once per project. Each use case records spatial boundary (field, farm, value chain, NUTS-level region, national, EU-macro), temporal boundary and resolution, functional unit, flows inside the boundary, effects treated as external (assessed as leakage rather than internalised), and whose costs, benefits and risks are counted. Boundary declarations are machine-readable metadata attached to every model run in the RBOS, so any result can be re-read with the assumptions that produced it.

Table 6 — Assumption register: test and failure response (reviewed at every gate; each assumption is linked to the critical-risk register, Table 3.1e)

ID	Assumption	How it is tested	Response if it fails	Linked risk
A1	Required regional and European data can be lawfully accessed, harmonised and reused	Asset-and-licence assessment (T4.1); per-pilot data-readiness check (T7.2) against the minimum viable dataset	Declared minimum viable dataset; proxy or lower-resolution substitute with pedigree downgrade; narrowed use case with reported limitation — never a silent scope reduction of a CR requirement (§3.1.6)	R2
A2	Farmers, accountants, businesses, authorities and researchers participate continuously	Participation, retention and influence monitoring by actor category (§1.2.12)	Reserve recruitment channels; adjusted formats, timing and compensation; rescoped co-design with reported representational limits	R9
A3	Analytical outputs are understandable, trusted and institutionally usable	Usability, comprehension and decision-relevance testing (T6.3, T8.3)	Simplify decision variables; add explanation and provenance views; shift from prescriptive to comparative outputs	R7
A4	Recommended pathways stay within food-security, ecological, safety and regulatory constraints	Admissibility gates applied before optimisation (§1.2.6)	Reject pathway, log failing gate and evidence, search the admissible set	R5
A5	Business models are accessible to a range of farm types and distribute value, cost and risk fairly	Profitability, transaction-cost, accessibility and fair-value assessment (T9.3)	Redesign configuration; test aggregation/cooperative variants; publish required enabling conditions	R6
A6	Policy and market conditions permit implementation, or conditional pathways can be defined	Policy monitoring and scenario analysis (T12.1, T5.1)	Convert the recommendation into a conditional pathway with explicit regulatory or market preconditions	R8

1.2.3 Governance framework: constructing and validating the IARGF

The IARGF specifies who decides what, on which evidence, subject to which safeguards, accountable to whom when agricultural resources are contested. It is developed in WP3 (T3.1–T3.4) as a co-designed artefact, not a literature synthesis.

T3.1 Governance, institutional and stakeholder assessment. (a) Systematic mapping of formal and informal decision arenas for land, biomass, water, carbon and biodiversity in each pilot territory, including competence, discretion and reporting duties; (b) policy-instrument analysis producing the **policy-to-requirement traceability matrix** (§1.1.6); (c) actor mapping with attributes (function, resource dependence, decision influence, information needs, capacity constraints); (d) semi-structured interviews and document analysis reconstructing how competing claims are currently reconciled. Sampling is purposive and stratified by actor category and territory; qualitative analysis uses an agreed codebook with inter-coder agreement checks and negative-case analysis. **[INPUT NEEDED: sampling frame, target numbers per pilot and actor category, coding software, coder arrangement, agreement threshold.]**

T3.2 Co-design of the framework. Facilitated co-design cycles in each hub plus cross-regional consolidation, working on concrete regional dilemmas rather than abstract principles. Each cycle produces a versioned framework release recording which actor input changed which rule (feedback-closure register, §1.2.12). Facilitation protocols, briefing materials and consent procedures are standardised across hubs. **[INPUT NEEDED: facilitation method, workshop cadence, participant numbers, compensation arrangements, consent templates.]**

T3.3 Safeguards and assessment framework. Actors and disciplinary experts jointly define, per context: safeguard variable; indicator; evidence required to demonstrate compliance; threshold or reference condition; evidence grade. Evidence classes are kept distinct and never treated as equivalent: **binding law**, **official reference condition or statistical standard**, **peer-reviewed scientific method**, and **project-negotiated provisional threshold** — the last always labelled as provisional, regionally negotiated and attributed. **[INPUT NEEDED: legal sources, threshold values, reference conditions, evidence-grading scheme.]**

T3.4 Decision rules, trade-off governance and fairness. The framework specifies the procedure for unavoidable conflicts: registration of competing claims; evidence that must be on the table; surfacing of distributional consequences; representation of minority and less-organised interests (tenants, small holdings, seasonal labour, downstream communities); documentation of decisions and dissent; revisiting when evidence changes. Where actor-elicited preference weights conflict, divergent weight sets are **reported side by side with their consequences for the ranking**, and minority weight sets are retained in the record; they are not averaged into a single set. Fairness is operationalised as distributive, procedural and recognitional, each with observable indicators. **[INPUT NEEDED: fairness indicator set and measurement instruments.]**

Validation. The IARGF is validated in the pilots against governance-effectiveness criteria: comprehensibility; feasibility within existing competences; ability to produce a defensible decision on a real regional dilemma; perceived legitimacy across actor categories; institutional fit. Releases are frozen at gate G3 before service integration and reopened only through documented change control. **[INPUT NEEDED: acceptance criteria, instruments, target values.]**

1.2.4 Evidence foundation: data, interoperability and the RBOS

The RBOS is the project's evidence infrastructure: an orchestration environment connecting data sources, models, analytical services and governance workflows so every result is reproducible and every input traceable. Functional boundaries are fixed to prevent duplicated development across WP4–WP6: **RBOS** provides data, metadata, provenance, model execution and service interfaces; **Engine** (WP5) provides scenario logic, gates, model coupling and comparison; **Services** (WP6) provide actor-facing framing, explanation and interaction. Each boundary has declared inputs, outputs, interfaces, owner and acceptance tests (gate G3).

T4.1 Requirements and asset assessment (reuse-first). A structured inventory of candidate assets — European and national datasets, Earth-observation services, land-use, soil, climate, hydrological and biodiversity data, farm accountancy and agricultural-economic accounts, market and price data, and existing models, tools and platforms from earlier Horizon Europe, LIFE and national programmes — is screened against a fixed protocol: relevance to a declared use case; coverage and resolution; documented quality and uncertainty; licence and access conditions; interface and format; maintenance status; interoperability effort. Each asset receives a recorded **reuse decision** (reuse as-is / adapt / interface only / do not use) with rationale and an explicit non-duplication statement (CR11). **[INPUT NEEDED: named datasets, models, tools, Earth-observation services and digital infrastructures with licences, coverage, resolution and update status; named Horizon Europe/LIFE projects; articulation with the Economic Accounts for Agriculture.]**

T4.2 Architecture, data governance and interoperability. Three interoperability levels: technical (interfaces, protocols, authentication, versioning, containerised model execution); semantic (controlled vocabularies, code lists, units, metadata profiles, mappings from source schemas to the project data model); organisational (roles, stewardship duties, data-sharing agreements, change control, service expectations). Governance controls cover provenance capture, quality flags, access control by data class, personal-data handling, retention and post-project stewardship (§2.2.1). **[INPUT NEEDED: standards, ontologies, metadata profiles, APIs, data-space alignment, conformance tests, security and cybersecurity measures, licences, service-level requirements.]**

T4.3 Integration and quality control. A documented pipeline — ingestion, validation against schema and range rules, harmonisation, gap treatment, aggregation to analysis units, quality scoring, publication of derived datasets with metadata. Two rules protect analytical honesty: **no silent imputation** (every filled gap flagged, method recorded) and **pedigree scoring** (each dataset carries a qualitative quality/uncertainty score that propagates into confidence statements on results). Integration is verified through interface tests, reproducibility tests (identical inputs and configuration reproduce identical outputs) and a **minimum viable dataset** per use case, below which an analysis is not run (gate G2). **[INPUT NEEDED: pedigree/quality-scoring scheme and thresholds.]**

Primary data. Pilot-generated data complement secondary sources: farm records and accountancy data; field or monitoring measurements where required; surveys and interviews with farmers, accountants, businesses and authorities; workshop outputs; service-interaction data from usability testing. All primary collection follows a common protocol with harmonised instruments, consent procedures, pseudonymisation and controlled access. **[INPUT NEEDED: measurement protocols, survey instruments, sampling frames, sample sizes with power or saturation rationale, recruitment procedures per pilot.]**

1.2.5 Analytical core: farm-to-macro modelling and the Decision Intelligence Engine

WP5 (T5.1–T5.3) turns CR6 into one operational capability rather than parallel disciplinary studies.

T5.1 Scenario architecture. Scenarios strictly separate external drivers (climate, market prices and demand, policy settings, regulatory change, technology availability) from controllable choices (land allocation, crop and rotation choices, residue and secondary-stream use, practice adoption, investment, contract and cooperation arrangements). Every analysis is a triple: **context scenario × pathway (choice bundle) × safeguard configuration**. This separates policy questions from decision questions, supports the price-threshold analysis of CR8 and enables cross-pilot comparison. **[INPUT NEEDED: scenario set, price and policy ranges, time horizons, pilot-level counterfactuals.]**

T5.2 Minimum coherent model chain and coupling. Fit-for-purpose components are coupled across scales, each retaining disciplinary integrity, and **only where a confirmed pilot decision requires them**: field/farm (biophysical response — yields, nutrient and organic-matter dynamics, water use, on-farm emissions and removals — plus farm economics: enterprise budgets, labour, capital, risk); farm decision (whole-farm allocation and investment modelling, formulated where justified as mixed-integer linear programming for indivisibilities, capacity, contract and rotation constraints, solved under multiple objectives); territorial/value chain (biomass supply–demand balancing, logistics and processing capacity, competing-use accounting, land-use allocation, landscape-level ecosystem-service and biodiversity response); regional/ macro (market and sectoral response, price-threshold behaviour, leakage beyond the territorial boundary); behavioural and institutional layer (adoption and participation representation from SSH evidence, §1.2.13, distinguishing technically optimal from plausibly adopted pathways). Components not required by a confirmed pilot decision are excluded at gate G1 and the exclusion is recorded.

Coupling is specified in a formal **coupling contract** per model pair: exchanged variables, units, spatial and temporal aggregation/disaggregation rules, direction of coupling, convergence criteria, uncertainty transmission. Aggregation and disaggregation errors are quantified and reported. **[INPUT NEEDED: model inventory with reuse status, equations, resolutions, calibration and validation datasets, coupling specifications.]**

Where AI is and is not used. The Engine is primarily an optimisation, simulation and structured-assessment system. Machine-learning components are used only where a specific justified need exists (gap-filling, pattern detection in monitoring data, surrogate modelling for large scenario ensembles), each documented with purpose, training and validation data, performance measure, uncertainty, bias testing, human-oversight arrangement and applicable legal requirements. **No component substitutes an opaque prediction for a safeguard test.** **[INPUT NEEDED: algorithms, training data, transparency, oversight, bias-testing and validation specifications per component, and applicable legal requirements.]**

T5.3 Comparison and ranking. Admissible pathways are compared using multi-objective optimisation to identify non-dominated sets, followed by multi-criteria assessment for actor-facing ranking. Three safeguards apply: (i) the objective set always spans economic, social, climate and biodiversity dimensions, any unquantifiable dimension carried forward as a documented qualitative criterion rather than dropped; (ii) weights are elicited from actors, recorded, versioned and varied in sensitivity analysis — never embedded as defaults; (iii) outputs present the **Pareto front plus rationale**, so users see what is sacrificed for what. **[INPUT NEEDED: objective functions, criteria set, normalisation and weighting-elicitation method, validation metrics.]**

Interdisciplinary integration mechanisms, all deliverable-bearing: shared glossary and indicator dictionary; **boundary objects** jointly populated by all disciplines (safeguard set, scenario protocol, indicator dictionary, coupling contracts); a named integration function accountable for cross-disciplinary consistency **[INPUT NEEDED: integration lead and responsible partner]**; mandatory cross-disciplinary internal review of every analytical deliverable, including at least one SSH and one biophysical reviewer; joint interpretation workshops where disciplinary teams and pilot actors reconcile divergent readings before publication. Unresolvable disagreement is published as documented divergence, not averaged away.

1.2.6 Safeguard-constrained decision logic

This is the project's central methodological commitment and the mechanism protecting food security, ecological integrity, safe circularity and legality in a project that also seeks farmer income and bioeconomy value (CR1, CR5, CR9). It is executable logic inside the Engine, exposed transparently in the services and governed by the IARGF. Sequence: define decision and boundary → Gates 1–5 in order → admissible set → comparison (§1.2.5) → reporting.

Table 7 — Admissibility gates

Gate	Test	Basis and instruments	Failure action
1 Food security (four dimensions)	Availability: effect on food and feed output from the territory and on the resource base underpinning future output. Accessibility: price, local supply, affordability. Utilisation: nutritional quality and food safety. Stability: exposure to shocks, seasonality, inter-annual variability	Each dimension has an indicator, data source and decision test. Test class is declared per indicator: **legal constraint**, **minimum standard**, **precautionary screen** or **comparative indicator**; only the first two can produce an automatic rejection, screens and indicators trigger mandatory documented review by the pilot board **[INPUT NEEDED: food-security methodology, indicators, data sources, thresholds, test classes, responsible owner]**	Reject or escalate to documented review; log reason, evidence and confidence
2 Ecological and natural-capital thresholds	Soil organic matter and erosion risk, nutrient balances, water quantity and quality, habitat extent and condition, landscape connectivity, greenhouse-gas balance including removals	Legal thresholds where they exist; official reference conditions otherwise; provisional negotiated thresholds labelled as such (§1.2.3). Indicators shared with the condition accounts (§1.2.7) **[INPUT NEEDED: indicator definitions, thresholds, reference conditions]**	Reject; log
3 Leakage, cumulative effects, conflicts of use	Pathways that improve local performance by displacing production, biomass demand or environmental pressure elsewhere; cumulative effects of multiple simultaneous pathways; water-use conflicts (irrigation, processing, ecological flows, domestic supply)	Declared leakage boundary and **materiality threshold** per pilot; uncertain or partially displaced effects are flagged and carried into the uncertainty statement rather than silently rejected or ignored **[INPUT NEEDED: leakage boundary definitions, materiality thresholds, cumulative-impact method, water-conflict assessment method]**	Reject if material; otherwise flag with condition
4 Safe circularity (biomass quality, food/ feed safety, traceability, regulatory suitability)	Risk-based screening of residues, secondary streams, waste-derived materials and underutilised biomass by intended end use (food, feed, materials, energy, soil return): contaminant and hazard profile, quality and homogeneity, seasonality, traceability, regulatory admissibility. Soil and nutrient functions are treated as competing legitimate uses: residue removal is admissible only when the soil/nutrient function it currently performs is accounted for	Not required by the topic text; included because pathways failing safety or regulatory screening cannot deliver the topic's outcomes **[INPUT NEEDED: hazard and screening protocols, applicable legal requirements, qualified safety/regulatory experts, acceptance criteria]**	Reject; log
5 Feasibility	Technology availability, processing and logistics capacity, contractual and permitting requirements, labour and skill availability	Pilot-specific feasibility evidence from WP7–WP9	Reject; log

Rejections are never discarded. Each carries the failing gate, the evidence used, the confidence level and the condition under which the pathway could become admissible. The **rejection log is a research output** and a direct input to policy recommendations (WP12).

Reporting. Every recommendation carries: gates passed with evidence grade; comparative performance across all four objective dimensions; uncertainty and robustness statement; distributional analysis (who gains, who bears cost and risk); and the explicit reason for the recommendation.

Figure 3 — Safeguard-constrained decision logic (placeholder, max 0.4 page): single graphic rendering of Table 7 as a gated sequence with rejection log, admissible set, Pareto/MCDA comparison and reporting outputs. **[INPUT NEEDED: figure production.]**

1.2.7 Natural-capital and ecosystem-service accounting (CR3)

Accounting is treated as a chain that must be complete to be useful, built in six documented steps.

1. **Accounting boundary and units** — accounting entity (farm, value chain, territory), asset classes, reporting period, and the articulation between project accounts, the Economic Accounts for Agriculture and farm-level accountancy practice **[INPUT NEEDED: accounting standard(s) adopted and exact articulation with EAA and farm accounts]**.
2. **Biophysical extent and condition accounts** — asset extent and condition measured or modelled using the indicators shared with Gate 2, so accounts and safeguards are one consistent system rather than two parallel ones.
3. **Ecosystem-service flow accounts** — services supplied by the agricultural system to identified beneficiaries, with explicit attention to who benefits at which scale.
4. **Valuation** — techniques appropriate to the decision use, with technique, assumptions, counterfactual and limitations stated per valued service; non-monetisable values retained as qualitative criteria (P2) and never dropped for tractability; results reported as ranges with sensitivity, never single point values **[INPUT NEEDED: valuation techniques, data sources, counterfactual specification]**.
5. **Pricing and remuneration use cases** — translation into instruments: payment or contract design, result- versus practice-based remuneration, verification requirements, and the transaction and verification costs imposed on farmers and buyers. Nature-market mechanisms, including carbon-removal or biodiversity-related instruments, are modelled **conditionally**: their contribution to farm income is counted only where regulatory basis, buyer, verification route and permanence/leakage conditions are specified **[INPUT NEEDED: applicable regulatory basis, instrument specifications, verification requirements]**.
6. **Decision integration** — accounts enter the Engine as condition thresholds (Gate 2), objective dimensions (comparison), revenue and cost lines (business-model assessment) and monitoring indicators (farmer service). **Double-counting controls** apply: the same flow may not simultaneously count as safeguard compliance, marketed unit and unpriced co-benefit without disclosure of the overlap.

Methodological risk is high (R3): valuation and accounting choices can drive results more strongly than the underlying biophysics. Mitigation is procedural — a pre-registered accounting-and-valuation protocol agreed before results are produced, independent methodological review (§1.2.11), and mandatory reporting under at least two valuation configurations. **[INPUT NEEDED: named natural-capital scientific lead, responsible partner, review arrangement.]**

1.2.8 Business models, markets and value-chain configurations (CR4, CR5, CR8)

WP9 (T9.1–T9.4), with WP5 and WP7/WP8, delivers analysis of existing and development of new farmer-centred business models and value chains, in four stages.

T9.1 Baseline analysis. Mapping per pilot territory of existing business models and value chains: market operators and functions; current and competing uses of relevant biomass and land; volumes, seasonality and quality; contract forms; price formation and transmission; margins and where value accrues; transaction, verification and compliance costs; financing sources and access barriers; farmers' current position and bargaining power. Regional resource baselines quantify **technical** and **sustainable** availability separately — the latter after Gates 2 and 4 — so no business model is built on biomass that is not actually available. **[INPUT NEEDED: existing models and value chains to be analysed; market operators, price datasets, demand evidence, transaction-cost and financing assumptions, all region- and use-case-specific.]**

T9.2 Co-design. New configurations are co-designed with farmers, accountants, cooperatives and businesses using a business-model canvas extended with natural-capital revenue and cost lines, safeguard-compliance requirements, verification burden and risk allocation. Configurations explicitly include cascading use hierarchies, non-extractive options (value created without removing biomass or degrading the asset) and aggregation models (cooperative, contract, platform) addressing the scale disadvantage of smaller holdings.

T9.3 Assessment. Each configuration is assessed for: profitability at farm and chain level under multiple price and policy scenarios, including **price thresholds** at which production choices switch (CR8); whole-farm rather than enterprise-only effects, using farm-accountancy logic and farm typologies; risk exposure and variability; **accessibility** (capital, skills, land tenure, scale, digital access, administrative burden); **fair-value distribution** along the chain against explicit criteria rather than assertion; demand-side plausibility, including consumer acceptance and willingness-to-pay evidence **only where a consumer-facing claim is part of the configuration**; and regulatory and safety admissibility. **[INPUT NEEDED: financial metrics, farm typologies, counterfactuals, fair-value criteria and formula, consumer-research methods, validation thresholds.]**

T9.4 Validation and transfer conditions. Configurations are validated in the pilots with the actors who would operate them against agreed acceptance criteria; each validated model is published with a **transfer-condition statement** — the market, institutional, biophysical and policy conditions under which it is expected to work, and those under which it is not. This is the methodological link to the replication roadmaps (WP12) and prevents context-specific results being presented as general prescriptions.

Farm accountants are a methodological asset, not only a mandated actor category (WP 2026–27, p. 187): they supply the accounting conventions used for income and natural-capital lines, test whether project metrics can be produced from data farms actually hold, quantify administrative and verification burden, and provide the advisory channel through which validated models reach farmers. **[INPUT NEEDED: farm-accountancy participant and its substantive resourced tasks.]**

1.2.9 From analysis to use: actor-oriented services and human-centred design (CR7)

T6.1 Requirements. For farmers, farm accountants and advisers, cooperatives, businesses, regional authorities and policymakers, the project documents: the decisions they actually make, at what moments, with what information and constraints; actionable decision variables; required outputs; data-entry tolerance; digital access and skills; trust requirements. Requirements are captured as user journeys with acceptance criteria signed off by actor representatives (gate G1).

T6.2 Design and integration. Services are built on RBOS–Engine integration with four design rules: decision-relevant framing (options and consequences in the user's terms — income, risk, workload, compliance, asset condition — not model diagnostics); explainability by construction (every recommendation can be opened to show gates applied, data and assumptions used, and what would change the result); uncertainty made usable (ranges, robustness and confidence in a tested visual grammar); accessibility (low-bandwidth and low-data-entry modes, plain language, localisation, adviser-mediated paths for users who will not interact with software). **[INPUT NEEDED: interface specifications, languages, accessibility standard, service-level requirements.]**

T6.3 Validation. Services are validated with real users on real decisions in the pilots via task-based usability testing, comprehension checks, decision-relevance assessment and adviser observation, against pre-agreed acceptance thresholds. Findings feed the refinement loop and are reported including failures. **[INPUT NEEDED: usability, comprehension, decision-relevance and acceptance targets; participant numbers per actor group.]**

1.2.10 Regional multi-actor pilots: design, common protocol and validation

WP7 prepares the pilots (T7.1–T7.3); WP8 implements, validates and refines (T8.1–T8.4).

Design principles. A **common core protocol** in every hub; **harmonised indicators** enabling cross-pilot comparison; **shared acceptance criteria**; **context-specific adaptation** of use cases to the regional challenge; **cross-pilot comparison** as an analytical objective in its own right; **iterative feedback, refinement and retesting** rather than one-shot demonstration.

Hub coverage. The concept envisages regional multi-actor pilot hubs spanning contrasting European biogeographic, structural and institutional contexts. **The number, identity, host, leader, sites, users, datasets and regional challenge of each pilot are [INPUT NEEDED: final pilot count, country, region, legal host with written confirmation of access to farms, records, monitoring sites, authorities and market operators; pilot leader; participating users; baseline datasets; regional challenge addressed].** No pilot-specific fact is asserted until confirmed, and no context-type list is presented as a commitment.

Common protocol (eight steps; regionally adapted content). (1) Hub establishment and actor recruitment against a required actor-mix specification; (2) regional challenge and use-case specification; (3) baseline consolidation and data-readiness verification against the minimum viable dataset; (4) counterfactual definition; (5) deployment of framework, services and configurations; (6) structured validation across the thirteen dimensions below; (7) feedback capture, component refinement and **retesting** of changed components; (8) regional consolidation into an action plan and governance roadmap. Progression from step 3 to step 5 is gated: a pilot does not deploy until data readiness (G2) and co-design sign-off and component verification (G3) are confirmed; deployment authorisation is G4 (Cycle 1) and G8 (Cycle 2). **[INPUT NEEDED: baseline/endline schedule, sample sizes with rationale, recruitment procedures, acceptance thresholds, data-quality and missing-data rules, refinement-and-retest procedure.]**

Validation dimensions (13), each with indicator, instrument, owner and acceptance criterion — to be completed in the indicator dictionary before G1: technical functionality; analytical validity and relevance; interoperability; usability and accessibility; governance effectiveness; food-security compliance; ecological and natural-capital performance; food/feed safety and regulatory suitability; economic feasibility; behavioural realism; stakeholder acceptance; institutional fit; transferability. **[INPUT NEEDED: indicator, instrument, owner and acceptance criterion for each of the 13 dimensions.]**

Cross-pilot analysis. A structured contextual-comparison design: identical protocol and indicators, systematically varying context (biophysical conditions, farm structure, institutional setting, market maturity, data availability). The question is not only does it work? but which contextual conditions explain differences in performance and acceptance? — which makes the replication roadmaps (T12.4) evidence-based rather than generic. Where pilot conditions prevent a clean counterfactual, effects are reported as **contextualised observations with named confounders**, never as attributed impacts.

Table 8 — Pilot design and common validation matrix (max 0.75 page; only differentiating facts are recorded per hub — common protocol, indicators and acceptance criteria are defined once)

Field (per hub)	Status
Hub identifier, country, region, NUTS level	[INPUT NEEDED, per confirmed hub]
Legal host and pilot leader; access rights confirmed	[INPUT NEEDED, per hub — written confirmation required before any pilot claim]
Regional challenge and use case(s)	[INPUT NEEDED, per hub]
Resource focus (biomass / land / water)	[INPUT NEEDED, per hub]
Actor mix (farmers / accountants / businesses / authorities / other), numbers per category	[INPUT NEEDED, per hub]
Baseline datasets, access route, minimum viable dataset met (G2)	[INPUT NEEDED, per hub]
Counterfactual or comparison design; named confounders	[INPUT NEEDED, per hub]
Scenarios tested; safeguard configuration (gate thresholds)	[INPUT NEEDED, per hub]
Business-model configurations tested	[INPUT NEEDED, per hub]
Retest scope after Cycle-1 refinement	[INPUT NEEDED, per hub]
Common to all hubs	Core protocol (8 steps, §1.2.10); 13 validation dimensions with common indicators **[INPUT NEEDED: indicator dictionary]** ; common acceptance criteria **[INPUT NEEDED: thresholds]** ; common consent, pseudonymisation and data-governance rules (§1.2.4)

1.2.11 Verification, validation, uncertainty and quality management

Three-level verification and validation. (1) Component verification: each dataset, model and service tested against its specification — schema and range tests, mass/energy/area balance checks, unit consistency, solver feasibility and optimality checks, code tests, reproducibility tests. (2) Integration verification: coupling contracts tested with benchmark cases; cross-scale consistency checks confirming that aggregated farm-level results are consistent with territorial results within declared tolerance; regression tests confirming that refinements do not break verified behaviour. (3) Validation in use: pilot testing across the thirteen dimensions (§1.2.10), including behavioural realism against observed actor behaviour and analytical relevance against actor judgement. Where empirical validation data are unavailable, the project performs and reports **structural validation** (documented expert appraisal of model logic, assumptions and behaviour) and labels the result accordingly; structurally validated results are never presented as empirically validated. **[INPUT NEEDED: model acceptance thresholds and validation datasets.]**

Independent methodological review is defined, not asserted: reviewers hold demonstrable competence in the reviewed domain, are not beneficiaries of the reviewed WP, review against published criteria at G3, G5 and G8, and issue findings that enter a tracked corrective-action procedure with a closure decision recorded in the milestone evidence. **[INPUT NEEDED: reviewer identification, independence rules, remuneration route (see Table 3.1g), corrective-action procedure.]**

Table 9 — Uncertainty classes, treatment and communication (uncertainty accompanies every result presented to any user; unquantifiable uncertainty is named rather than omitted)

Class	Source	Treatment	Communicated as
Input/ data	Measurement error, coverage gaps, resolution mismatch, imputation	Quality flags, pedigree scoring, propagation via Monte Carlo or bounded intervals	Data-confidence label on each result
Parametric	Uncertain coefficients, prices, cost and yield parameters	Screening sensitivity analysis, then variance-based sensitivity where feasible	Ranges; ranked list of influential parameters
Structural	Model form, coupling simplifications, aggregation choices	Alternative model or specification runs; documented structural-validation review	Explicit "results depend on model form" statement with alternatives shown
Scenario	Future prices, policy, climate, demand	Scenario ensembles; no probability attached to scenarios	Robustness across scenarios; regret/sat